INCH POUND

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DETAIL SPECIFICATION

REEL, CABLE RC-435( ),U

INACTIVE FOR NEW DESIGN AFTER 25 FEBRUARY 1998

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. This specification covers one type of tubular steel reel designated as Reel, Cable, RC-435( )/U (see 6.3).

2. APPLI CABLE DOCUMENTS

2. 1 General. The documents listed in this section are specified in section 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-10578 - Corrosion Removing and Metal Conditioning Compound (Phosphoric Acid Base)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-13231 - Markings of Electronic Items

(Copies of these documents are available online at https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue Philadelphia, PA 19111-5096 or e-mail to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY COMMUNICATIONS & ELECTRONICS MATERIEL READINESS COMMAND DRAWINGS

SC-DL-69296 - Reel, Cable RC-435( )/U, Drawing and Data list

(Copies of these documents are available from US Army Communications & Electronics Materiel Readiness Command Logistics Engineering Dir. 3200 Raritan Ave., Aberdeen Proving Ground, MD 21005 or https://cecom.army.mil.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from www.asq.org or the American Society for Quality, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The reel which this specification covers is Reel, Cable RC-435( )/U which is used for storage and transportation of various types of wire, cable or cable assemblies.

3.2 Construction. The reel shall be constructed in accordance with the requirements of this specification and Drawing and Data List SC-DL-69296.

3.3 Welding. All welds shall be of ample size with thorough penetration, good fusion and stronger than the material which they join.

3.3.1 Cleaning prior to welding. Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.3.2 Process. Preheating shall be employed where distortion is likely to result from welding. Welds shall have a thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved.

3.3.3 Cleaning after welding. Welded assembles shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to finishing steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with MIL-C-10578. Acid used for cleaning shall be completely neutralized and removed. Sandblasted items shall be finished or a protective coating shall be applied within four hours of cleaning.

3.4 Finish, protective. The reel shall be given protective finish in accordance with equipment drawings. Finish shall be smooth and free from blowholes (see 4.4). Finish shall be applied after satisfactory completion of tests specified in 4.6, 4.7, and 4.8.

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3.5 Identification marking. All reels shall bear identification markings in accordance with the applicable detailed drawing and MIL-STD-13231 (see 3.2).

3.6 Hub size. When tested as specified in 4.5, the axle shall pass through the hub without binding.

3.7 Alignment. When tested as specified in 4.6, the rim shall be no more than one-fourth of an inch from the perpendicular to the axle.

3.8 Drop. The reel shall show no evidence of permanent degradation (other than deformation) and no loosening of parts as a result of the drop test specified in 4.7. The permanent deformation shall not exceed by more than one-half inch the measurements made in 4.6.

3.9 Vibration. When tested as specified in 4.8, the reels shall show no evidence of permanent degradation (including no permanent deformation or loosening of parts)..

3.10 Moisture resistance. Upon completion of test specified in 4.9, there shall be no evidence of corrosion or any other form of deterioration.

3.11 Workmanship. The reels shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein (see 4.4):

Construction	3.2
Welding	3.3
Cleaning prior to welding	3.3.1
Process	3.3.2
Cleaning after welding	3.3.3
Finish	3.4
Marking	3.5

3.12 First article. When specified in the contract, the contractor shall subject samples to first article inspection in accordance with paragraph 4.2.

3.13 Mechanical safety. Exposed edges shall be rounded sufficiently to minimize laceration/puncture hazards, the goal being a minimum radius of 0.04 inch (1 mm).

4. VERIFICATION

4.1 Classification of inspections. The inspections requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. This inspection shall consist of the inspection specified in Table I and shall include, as referenced on inspection data sheets, inspection specified in subsidiary documents and the supplementary tests. Order and quantity of equipment to be subjected to first article testing shall be as specified in Table I.

4.2.1 Standard of workmanship. Standard of workmanship, 3.11, shall be established during the visual inspection of 4.4.

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TABLE I. First Article Test Schedule.

Title	Requirement paragraph	Test paragraph	Sequence	
			#1	#2
Workmanship	3.11	4.4	1	1
Hub size	3.6	4.5	3	
Alignment	3.7	4.6	2	2
Drop	3.8	4.7		3
Vibration	3.9	4.8		4
Moisture Resistance	3.10	4.9	4	

4.3 Conformance inspection. The inspection specified in 4.2 and 4.3.1 through 4.3.3 shall be performed to demonstrate compliance with specified requirements..

4.3.1 Group A inspection.

4.3.1.1 Inspection lot. An inspection lot shall consist of all reels of the same type, produced under essentially the same conditions and offered for inspection at one time.

4.3.1.1 Group A tests. Group A tests shall consist of the tests specified in Table II, and shall be made on the same set of sample units in the order shown.

4.3.1.1.1 Sampling plan. Statistical sampling and inspection shall be in accordance with ASQ Z1.4 for general inspection level II.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test Method paragraph
Workmanship	3.11	4.4
Hub size	3.6	4.5

4.3.2 Group B inspection. This inspection, including sampling, shall conform to Table III and to the special procedure for small-sample inspection of ASQ Z1.4. Group B inspection shall be performed on inspection lots that have passed Group A inspection.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test Method paragraph
Alignment	3.7	4.6

4.3.3 Group C inspection. The inspection shall consist of the tests specified in Table IV, and shall be performed on samples that have been subjected to and met Group A and Group B inspection. Sample units shall be selected at random without regard to their quality except that the samples selected at the start of production shall be selected from the first units produced. The tests specified in Table IV shall be performed at the beginning of production and according to the frequency specified, (months or units) whichever comes first.

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TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph	Frequency Months	Frequency Units
Drop	3.8	4.7	1	1,000
Vibration	3.9	4.8	3	2,500
Moisture Resistance	3.10	4.9	5	5,000

4.4 Workmanship. The reels shall be given a thorough visual and mechanical inspection to ascertain that the materials, construction, identification marking, finish, and workmanship are in accordance with applicable requirements (see 3.11).

4.5 Hub size. After finish a 1-1/32 +0 -1/64 by 1-1/32 +0 -1/64 inch axle shall be inserted through the reel hub at two angles so that all grooves are tested to insure compliance with 3.6.

4.6 Alignment. Prior to applying finish the requirements of 3.7 shall be met when a 1-1/32 +0 -1/64 by 1-1/32 +0 -1/64 inch axle is inserted into the hub of the reel. Measurements shall be made at four points 90° apart on each rim.

4.7 Drop. The reel prior to finishing shall be loaded with a weight and quantity of cable comparable to its normal use as follows: 55 to 65 pounds including weight of reel and loaded to 80 to 90% of capacity.

4.7.1 The reel shall be raised by a hoist equipped with a “quick release” device, to a height of 3-1/2 feet above a platform built of two inch fir backed by a concrete floor. By tripping the “quick release” device the reel shall be dropped squarely on the periphery of both rims.

4.7.2 The reel shall then be raised to a height of 3-1/2 feet and held so that when dropped, it will land squarely on one side. Then repeat on the other side.

4.7.3 The reel shall then be raised to a height of 3-1/2 feet and canted so that one rim is more than one inch higher than the other, so that when dropped it will land at an angle to the periphery of one rim but in an area opposite the opening on each rim. Then repeat on the other side.

4.7.4 At the conclusion of the tests, the reels shall be examined for compliance with 3.8.

4.8 Vibration. Prior to applying finish, the reels shall be subjected to the tests of 4.8.1 thru 4.8.4. During vibration the unit shall be clamped in its normal operating position on a vibration table that can be controlled to within 10 percent of the specified frequencies and amplitudes.

4.8.1 Direction of vibration. The reels shall be vibrated successively, over the specified ranges of applied frequencies, in three mutually perpendicular directions parallel to the edges of the reels.

4.8.2 Rate of change of vibration frequency. The frequency of the applied vibration shall be varied uniformly at a rate of approximately one Hz. The entire frequency range, from 10 to 33 cps and return to 10 cps, shall be traversed in approximately one minute.

4.8.3 Amplitude and frequency of applied vibration. The amplitude (one-half of the total excursion) of the vibration table and the frequency range shall be in accordance with the following:

Frequency of vibration (Hz)	Amplitude of vibration (in)	
	Horizontal	Vertical
10 to 33	0.020	0.030

4.8.4 Duration of test. The reels shall be vibrated for at least 90 minutes, 30 minutes in each of the three directions specified in 4.8.1.

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4.8.5 At the conclusion of the test, the reels shall be examined for compliance with 3.9.

4.9 Moisture Resistance. The reel shall be subjected to fifteen continuous 48 hour moisture resistance cycles. A single cycle shall consist of Steps 1 through 8. The relative humidity shall be maintained between 90 and 98 percent at all times throughout each cycle and during the first conditioning period. The actual temperature of the conditioning periods and during each cycle at steps 2, 4, 6 and 8 shall be within 3 degrees F of the temperature specified.

Condition the reel at 86 degrees F for 24 hours before test.

Step 1. Raise the internal chamber temperature from 86 to 149 degrees F at a rate of not less than 15 degrees F per hour.

Step 2. Maintain the temperature at 149 degrees F for 8 hours.

Step 3. Lower the temperature from 149 to 86 degrees F at a rate of not less than 15 degrees F per hour.

Step 4. Maintain the temperature at 86 degrees F for 20 hours.

Step 5. Lower the temperature from 86 to 68 degrees F within 2 hours.

Step 6. Maintain the temperature at 68 degrees F for 4 hours.

Step 7. Raise the temperature from 68 to 86 degrees F within 2 hours.

Step 8. Maintain the temperature at 86 degrees F for 4 hours.

After completion of the fifteen cycles, condition the reel at 73 degrees F and between 40 and 60 percent relative humidity for 24 hours. Then inspect the reel for any evidence of corrosion or other deterioration. The reel shall meet the requirements of 3.10.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from then managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The reel covered by this specification is intended for use in storing and transporting wire, cable or cable assemblies.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of the specification.
b. Type required.
c. Place of final inspection.
d. Packaging requirements (see 5.1).

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6.3 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example, Reel, Cable RC-435W/U. As soon as possible after the award of the contract, the contractor should apply to the government office specified in the contract for such information (see 1.1).

6.4 Subject term (key word) listing.

Alignment
Finish, Protective
Hub Size
Mechanical Safety
Moisture Resistance
Tubular Steel Reel
Vibration
Welding

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
DLA - CQ

Preparing activity:
DLA - CQ

(Project 8130-2025-001)

Review activities:
Army - CR4

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